

System Technical Reference

AICONSUMERAGENT V1.2 • ENGINEERING DOCUMENTATION

1. Infrastructure Overview

AIConsumerAgent is a multi-process Electron application designed for privacy-first, self-hosted WhatsApp automation. Unlike cloud-based SaaS solutions, the application runs entirely on your local hardware, keeping all Signal Protocol keys and SQLite databases under your direct control.

1.1 Compute Layer

The system utilizes **Electron IPC (Inter-Process Communication)** to separate heavy LLM reasoning and WhatsApp bridge activities. The **Main Process** handles raw network sockets and WhatsApp Signal sessions via **Baileys**, while the **Renderer Process** manages the React-based UI and the asynchronous Agent Runtime.

1.2 Persistence Layer

Persistence is handled by **SQLite 3** with **FTS5 (Full-Text Search)** enabled. All conversation histories and learned entities are stored locally. For high-availability auditing, the system mirrors state to **JSON audit logs** located in the **Library/Application Support** directory.

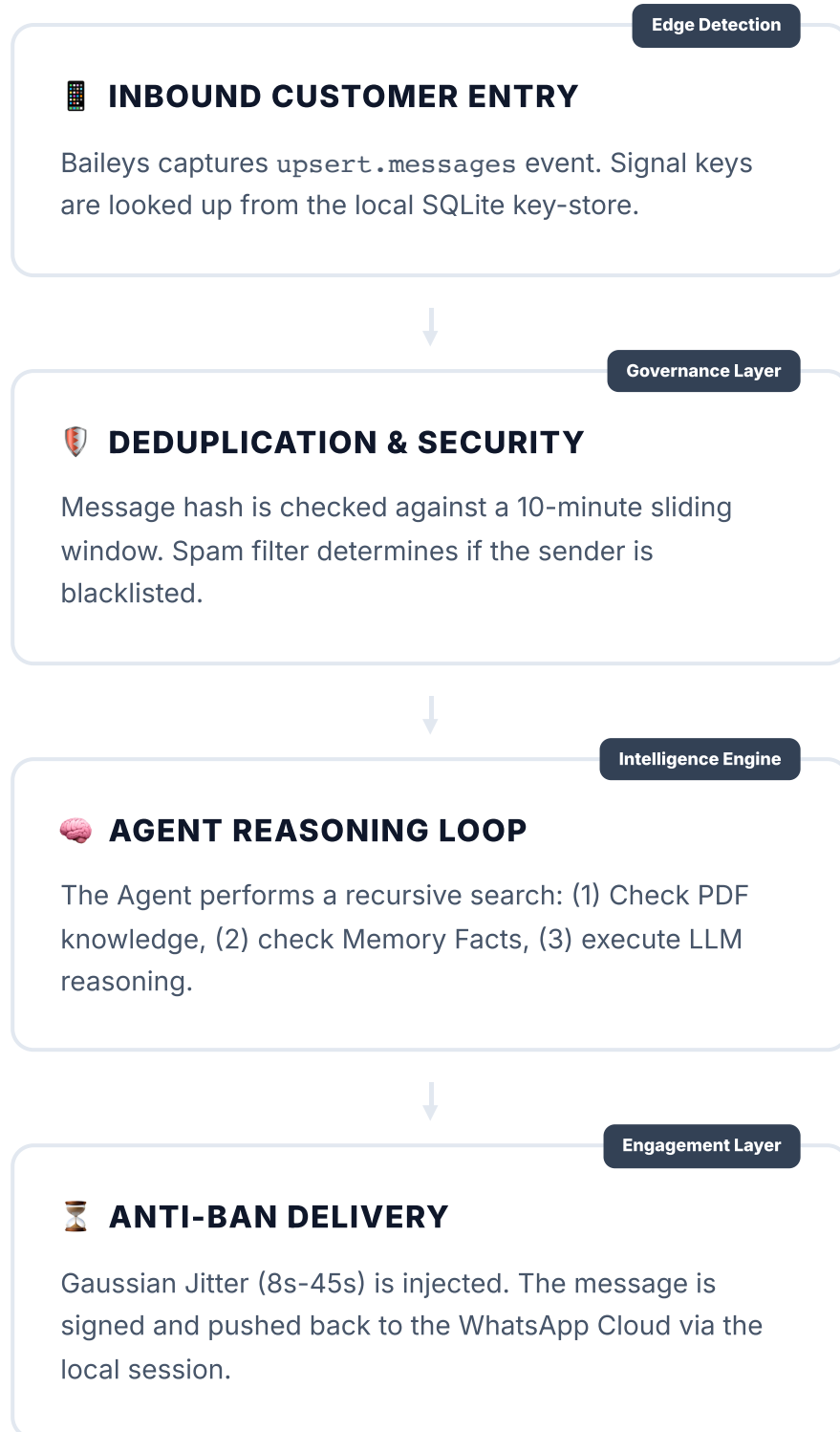
2. Feature Maturity Matrix

As a tester, it is critical to distinguish between verified production logic and experimental beta features. Use the matrix below to guide your evaluation.

COMPONENT	STATUS	FUNCTIONALITY DETAILS
WhatsApp Bridge	 Production	Full multi-device sync, QR pairing, and self-healing session management.
Vector RAG Engine	 Production	Local document ingestion (PDF/TXT), FTS5 semantic search, and context injection.
Admin Relay	 Production	Real-time escalation notifications to the admin's personal phone via linking.
Memory Facts	 Beta	Long-term entity extraction. Works but requires significant history to stabilize.
Dashboard Stats	 Beta	Core KPIs are real; "AI Topics" requires a manual sync button click to update.
Browser MCP	 Beta	Playwright automation. Requires local Chromium/Chrome installation.
Social Bridges	 Roadmap	Instagram, Telegram, and SMS bridges are UI-placeholders for Phase 2.

3. Detailed Architecture Flowchart

The following diagram outlines the path of a customer message from the Edge to the Reasoning Engine.



💡 Developer Note for Testers

If you encounter "Double Messaging," please check if your linked phone has a weak data

connection. The app uses an optimistic sync mechanism that can occasionally trigger a retry if the cloud ACK is delayed.

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